The Emergence of Prime Distribution from Low-Dimensional Deterministic Chaos

Bridging Number Theory and Nonlinear Dynamics through the 'Weak Chaos' Universality Class

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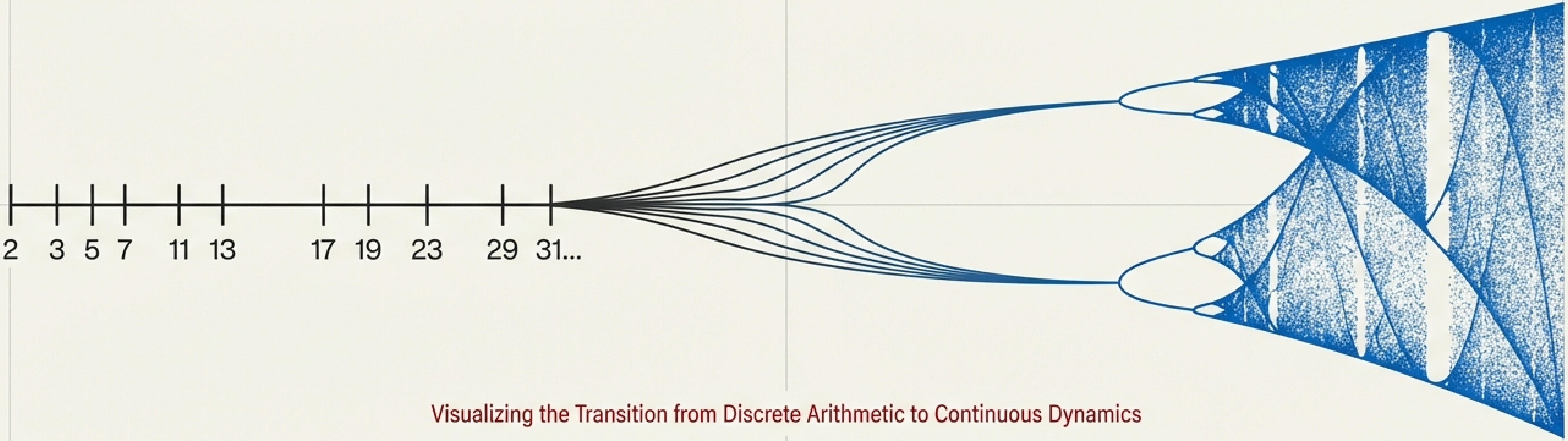
Discrete Arithmetic (Prime Numbers)

Continuous Dynamics (Bifurcation Diagram)



Visualizing the Transition from Discrete Arithmetic to Continuous Dynamics

A proposal that the "randomness" of prime numbers is not stochastic noise, but the trajectory of a deterministic system evolving at the Edge of Chaos.

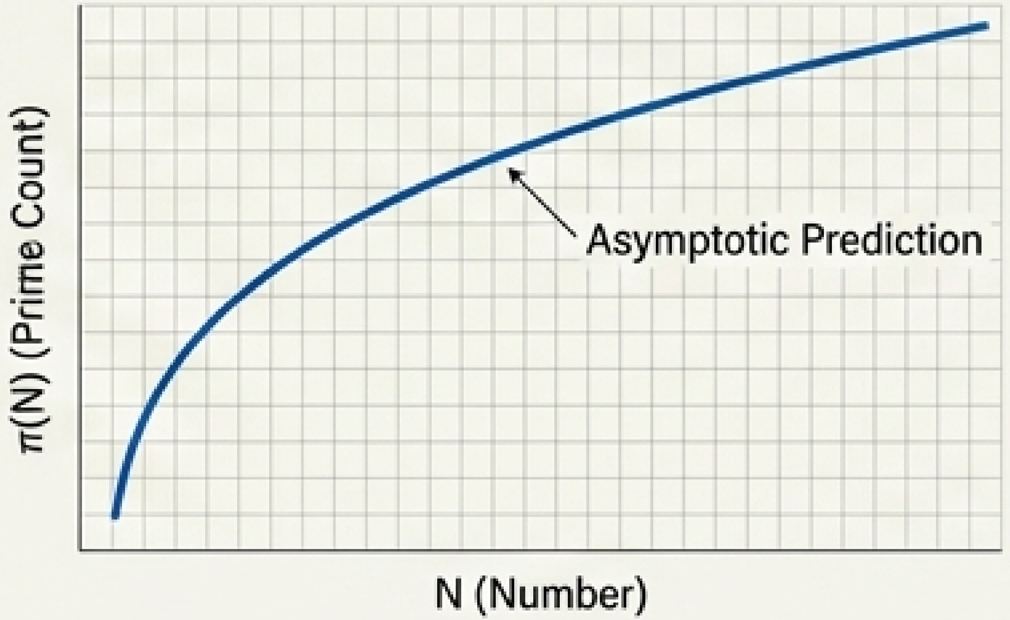
The Density Paradox

Macroscopic Randomness vs. Microscopic Rigidity

Macroscopic View (Random)

Since Riemann, primes are modeled as pseudo-random events. The Prime Number Theorem predicts a smooth asymptotic density:

$$\pi(N) \sim N/\ln N$$

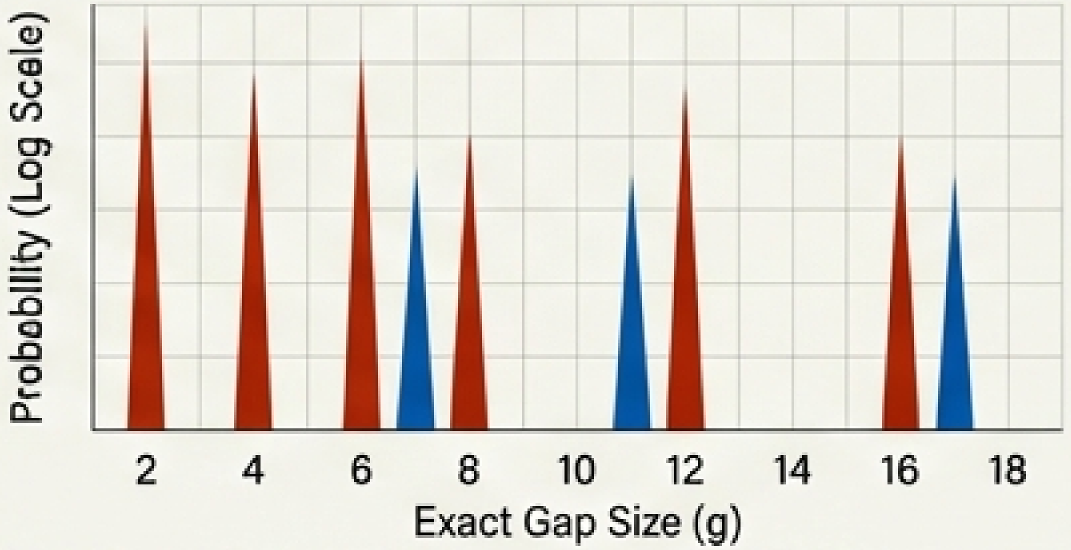


Probabilistic models successfully predict the *count*.

Microscopic View (Rigid)

However, primes exhibit strict arithmetic constraints indistinguishable from modular rules.

- **Parity Rigidity**: Two primes ($p > 2$) can never be consecutive. Gap ≥ 2.
- **Short-Range Correlations**: Primes exhibit 'exclusion zones' based on modular arithmetic.



Probabilistic models fail to explain the *structure*.

Probabilistic models can predict the *count*, but they cannot explain the *structure* of where primes appear.

The Failure of Pure Chance (The Cramér Model)

Why the ‘Random Dice Throw’ approach is insufficient.

Feature Dimension	Traditional Stochastic Model (Cramér)	Proposed Chaos Model
Core Mechanism	Random Dice Throwing	Deterministic Chaotic Orbit
Parity Rigidity	None (Allows Odd Gaps) ❌	Spontaneously Emergent (Gap ≥ 2) ✅
Memory Properties	No Memory (Lag-1 = 0) ❌	Short-range Repulsion (Lag-1 < 0) ✅
Dynamic Class	White Noise ($\lambda \to \infty$)	Weak Chaos ($\lambda \approx 0.1$)

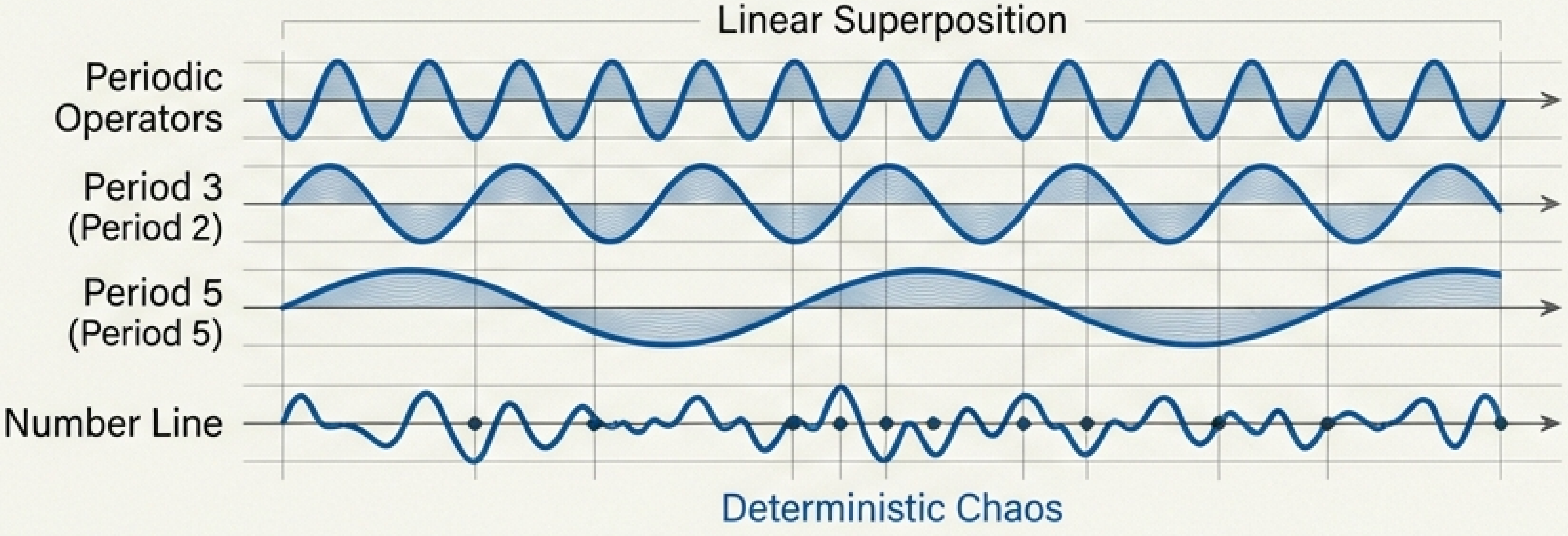
Treating primes as white noise ignores the fundamental ‘Parity Rigidity’ and memory inherent to the sequence.

The Paradigm Shift

From Stochastic Process to Deterministic Chaos

- **Weak Chaos**: The unpredictability of primes is not high-dimensional noise, but the signature of a simple nonlinear feedback loop.
- We reconceptualize the number line not as a static set, but as a medium for wave interference.

The Physical Metaphor



“The distribution of primes is the trajectory of a deterministic system evolving at the ‘Edge of Chaos’.”

The Dynamical Sieve Isomorphism

Mapping Arithmetic to Physics

1. Dynamic Sieve Process

Natural Numbers (N):

0 1 2 3 4 5 6...

Sieve out 2 (Remove 2k):

R L R L R L R... (Period 2)

Sieve out 3 (Remove 3k):

R L R R R L R... (Period 6)

→ Abstraction →

2. Symbolic Dynamics

Study Object:

RLRRRL...

(Periodic/Chaotic Seq.)

Kneading Theory links Sequence to Orbit.

→ 1-to-1 Mapping →

3. Orbit Dynamics

Logistic Map Equation:

$$X_{n+1} = 1 - uX_n^2$$

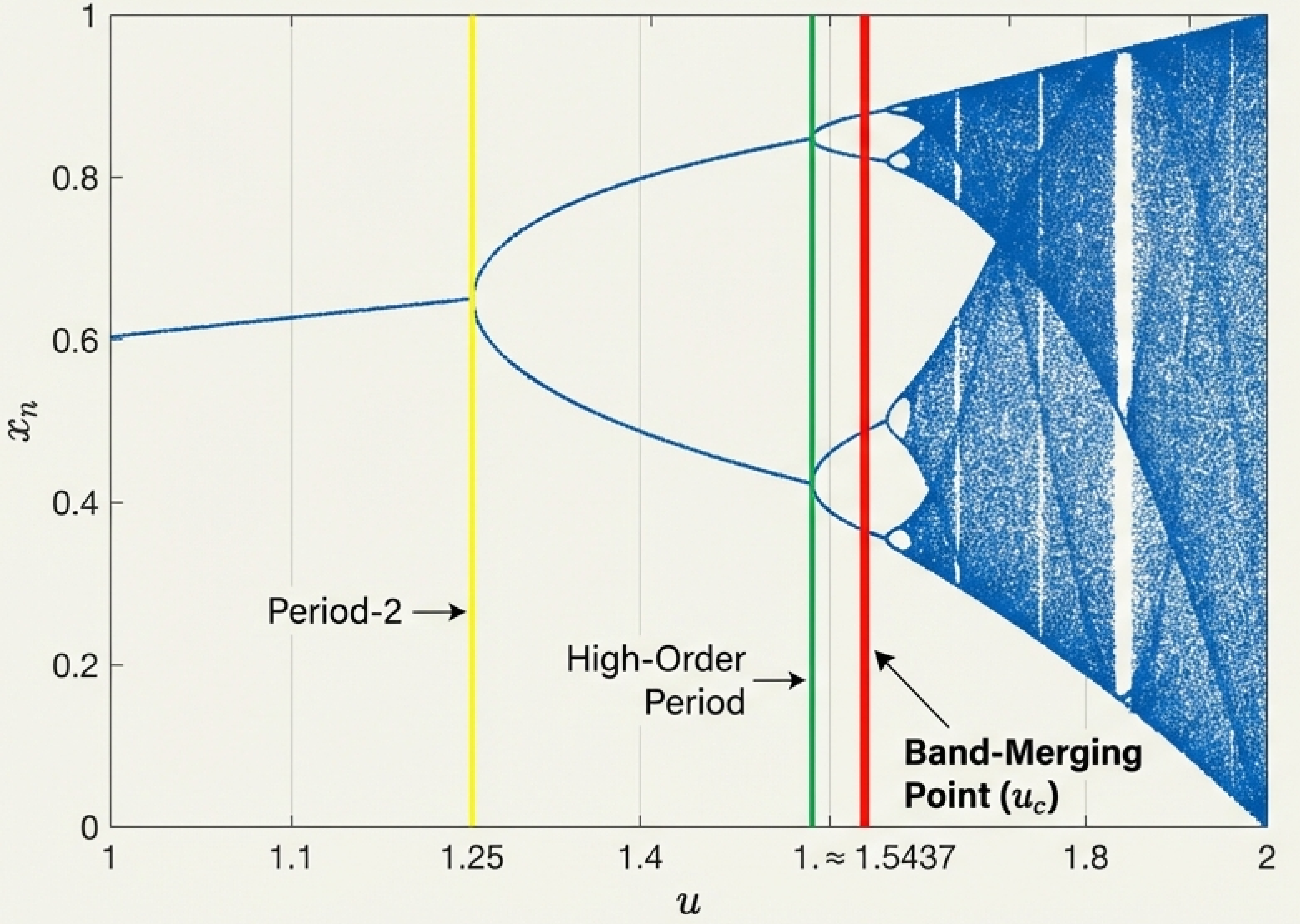
Conclusion (MSS Theory):

Sequence & Parameter u **Strictly Correspond**

The limit symbolic sequence Q_∞ of the Sieve is topologically isomorphic to the kneading sequence of the Logistic map.

The Edge of Chaos: The Band-Merging Point

Locating the Prime Distribution in the Bifurcation Diagram



The Critical Parameter: $u_c \approx 1.543689$

Why here?

This is the threshold of **Global Ergodicity** (the orbit explores the whole interval) while retaining **Local Rigidity** (the orbit remembers the period-2 bifurcation).

The parameter u_c represents the only topological state that balances global spread with strict local constraints.

Parity Rigidity & The Topological Skeleton

The Problem

In probability theory, a gap of 1 between primes (e.g., consecutive integers being prime) is merely “unlikely”.

In the Sieve, it is impossible (**Parity**).

The Solution: Topological Prohibition

The attractor at u_c retains “memory” of the Period-2 bifurcation.

The symbolic dynamics forbid the substring **LL** in the asymptotic limit.

The RLR Skeleton: The fundamental grammar of the orbit is RLR^{∞}. Parity is not an external rule; it is an intrinsic Renormalization Group Fixed Point.

The Aging Universe: Resolving the Density Problem

Why prime density decreases while the structure remains rigid.

$$\underbrace{u_n}_{\text{Time-dependent Parameter}} = \underbrace{u_c}_{\substack{\text{The Attractor}\\\text{(Structure/Parity)}}} - \underbrace{k(\ln n)^{-2}}_{\substack{\text{Thermodynamic Dissipation}\\\text{(Macroscopic Density)}}}$$

The Conflict: A static attractor implies constant prime density. But primes fade as $1/\ln n$.

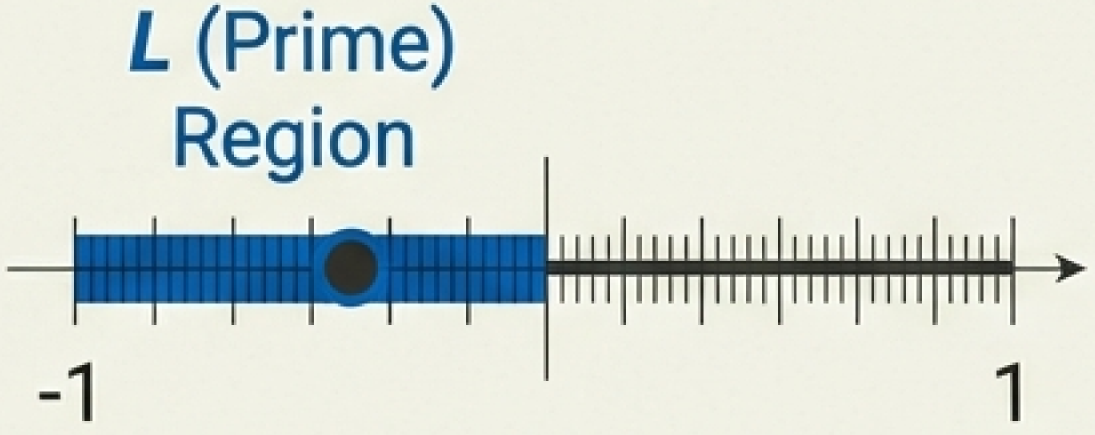
The Solution: The Non-Autonomous 'Aging' Hypothesis. The system undergoes slow parameter decay, dissipating 'energy' over time.

This decoupling allows us to capture the asymptotic flow without altering the fundamental arithmetic skeleton.

Dynamical Formulation of the Twin Prime Event

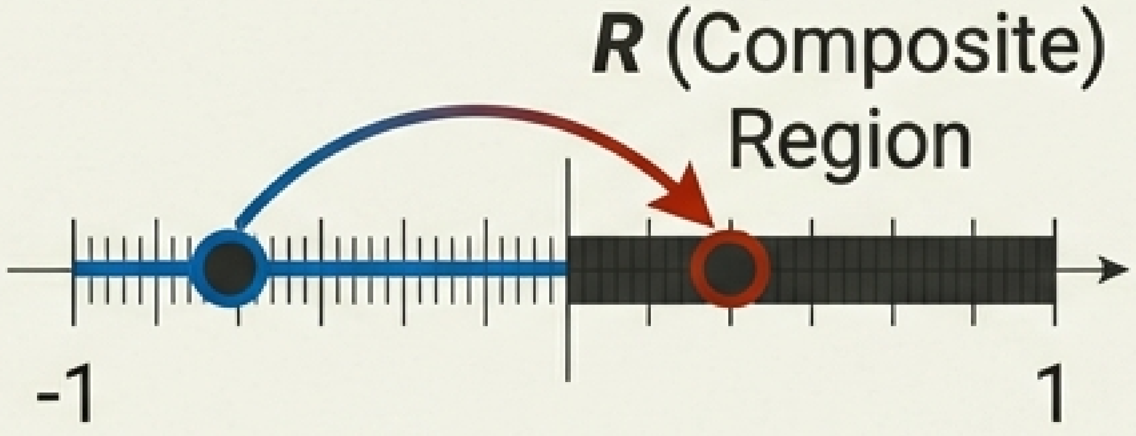
Defining a Twin Prime in Phase Space

State n (p)



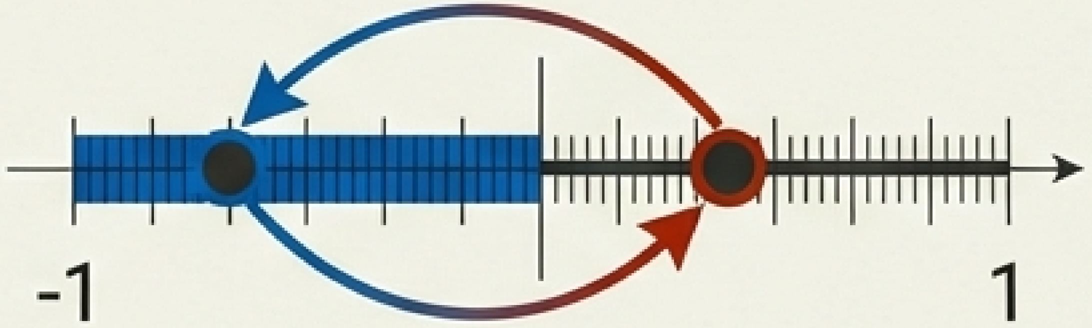
Orbit visits Prime Region.

State $n+1$ $(p+1)$



Orbit *must* visit Composite due to parity.

State $n+2$ $(p+2)$



Orbit returns to Prime Region.

The Cylinder Set: This sequence corresponds to visiting the geometric set I_{LRL} in phase space.

Deriving the Twin Prime Constant requires calculating the measure of this specific set under the influence of Aging.

Deriving the Twin Prime Constant (C_2)

An Analytical Solution

$$k \cdot \mu_{LRL} = 2C_2$$

k	μ_{LRL}	C_2
System's Aging Rate (Dissipation)	Intrinsic Geometric Probability (Calculated from Cylinder Set)	Structural Constant (Number Theory)
≈ 12.73	≈ 0.1037	≈ 0.66016

Result: The Twin Prime Constant is physically determined by the ratio of the system's aging rate to its intrinsic geometry.

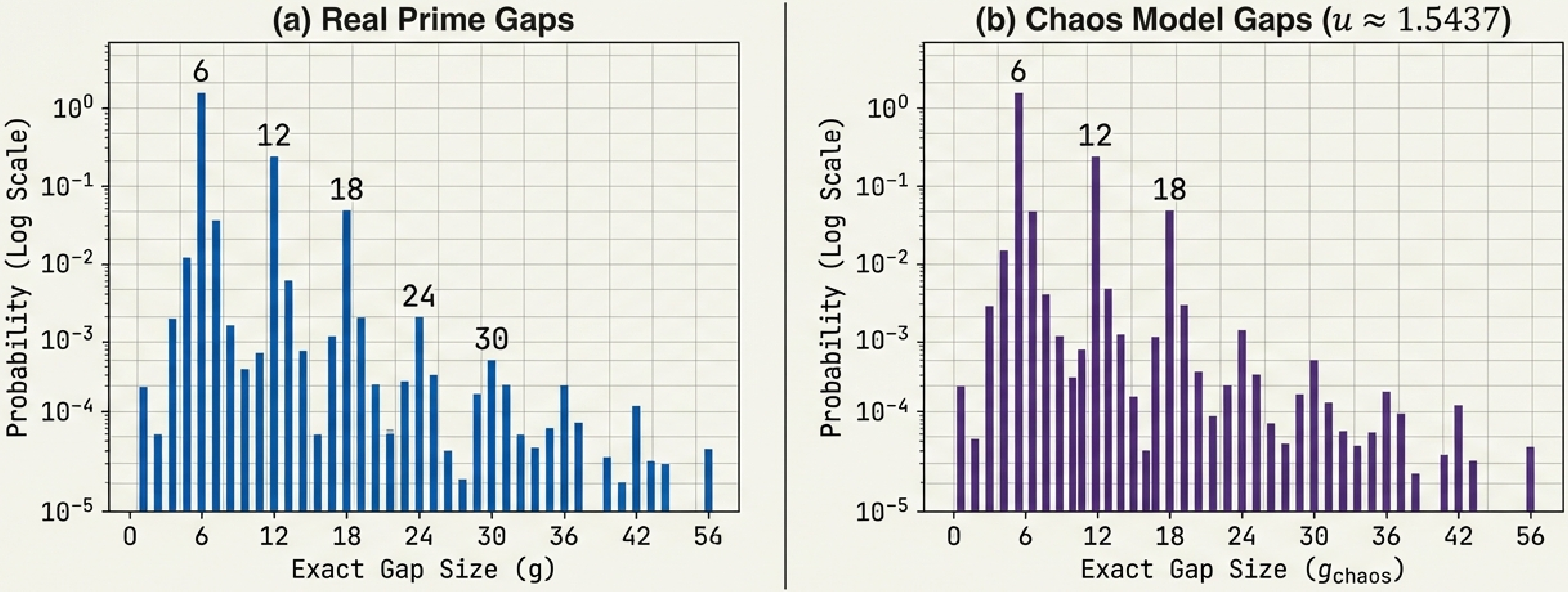
Quantitative Evidence: Precision Convergence

Precision $|C_{\mathrm{model}} - C_2| < 10^{-4}$.

This is not a curve fit. The constant emerges naturally as a fixed point of the dynamical system.

Microscopic Evidence: The Gap Spectrum

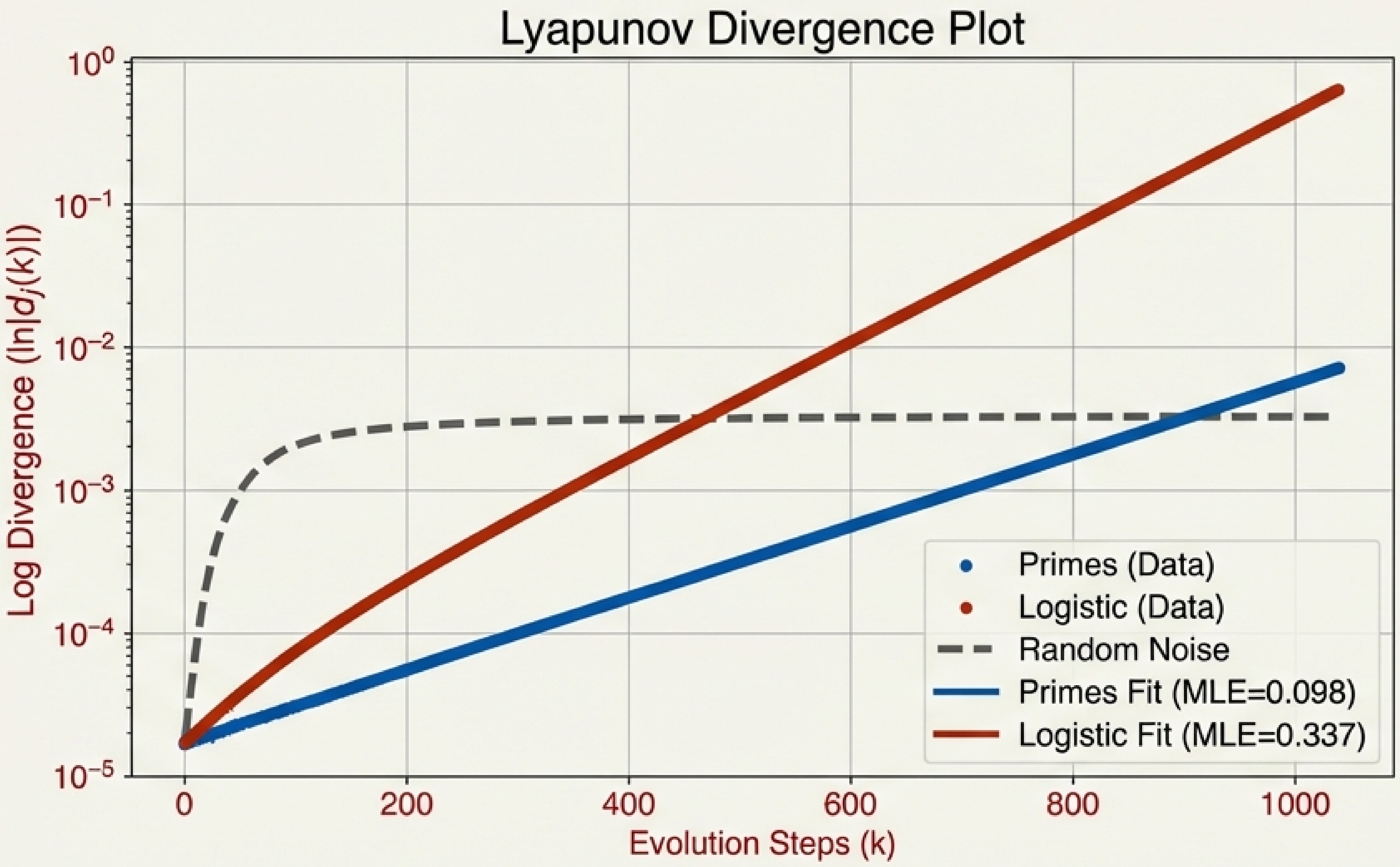
Does the model look like primes up close?



The Chaos model spontaneously reproduces the 'Needle Spectrum' and modular peaks purely from the $L - R - L$ kneading mechanism. Correlation > 0.99.

Quantifying the Chaos

Lyapunov Exponents & Entropy



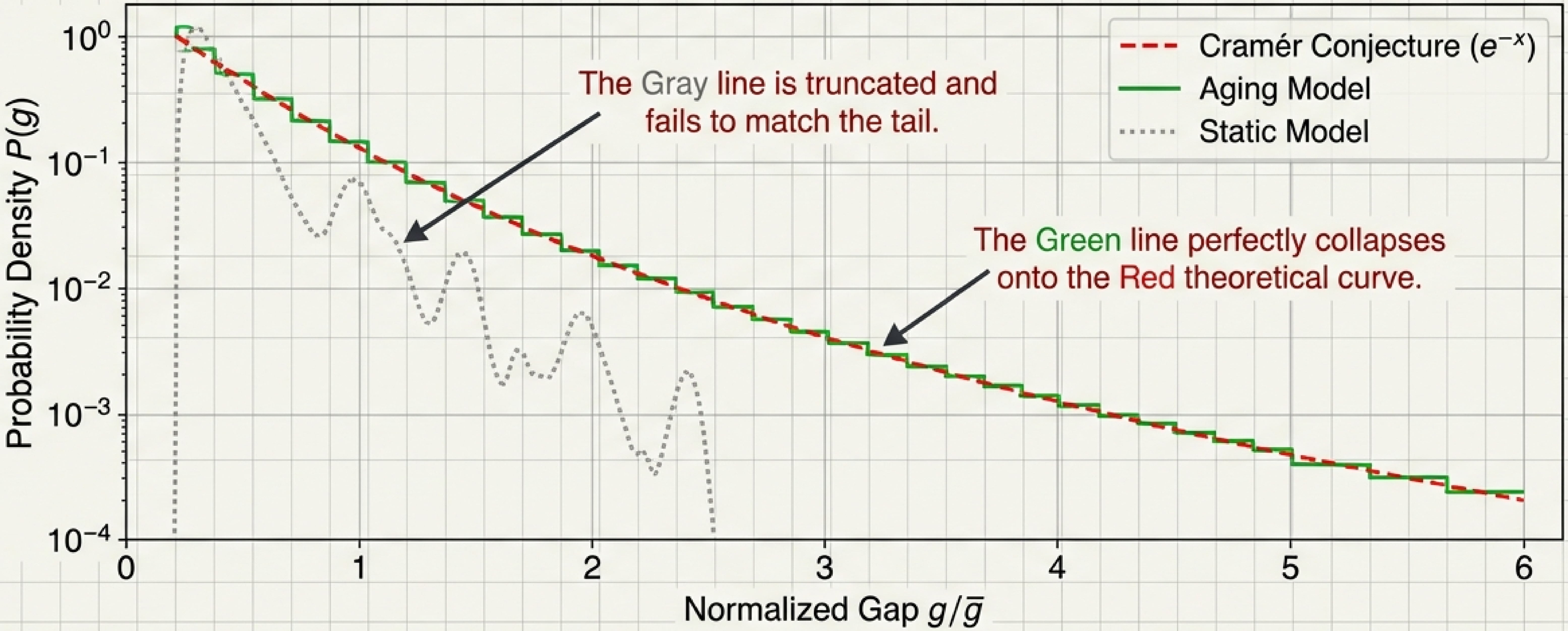
Maximal Lyapunov Exponent (MLE): $\lambda \approx 0.1$.

Positive (Unpredictable) but Small (Highly Constrained).

Conclusion: Primes belong to the 'Weak Chaos' Universality Class, balancing entropy generation with structure preservation.

Recovering Statistical Laws

The Cramér Conjecture



The 'Aging' mechanism successfully simulates the unbounded rarification of events, making the deterministic system statistically indistinguishable from a Poisson Point Process at large scales.

Conclusion & Implications

The Journey

- From Coin Tossing → Deterministic Orbit.
- Microscopic: Rigid/Deterministic (u_c).
- Macroscopic: Dissipative/Entropic (u_n).

The Riemann Hypothesis

- Suggests zeros of the Zeta function relate to eigenvalues of this specific chaotic operator.
- The "Needle Spectrum" links dynamical frequencies to prime periods.

Final Thought

Primes are not random; they are complex. Their distribution is the shadow of low-dimensional deterministic chaos.